

OOPS TEST

Q1: Write a program that takes a list of numbers and returns the sum of even numbers and the product of odd numbers.

Example input: [1, 2, 3, 4, 5, 6]

Expected output: Even sum: 12, Odd product: 15

Q2: Create a function that checks if a string is a palindrome (reads the same forwards and backwards).

Q3: Create a **BankAccount** class with:

- Attributes: account_number, account_holder, balance
- Methods: deposit(), withdraw(), get_balance(), display_account_info()
- Ensure withdraw doesn't allow negative balance

Q4: Create a base class **Vehicle** with:

- Attributes: make, model, year, color
- Methods: start(), stop(), display_info()

Create a derived class **Car** that inherits from **Vehicle** and adds:

- Attributes: number_of_doors, fuel_type
- Methods: horn(), open_fuel()

Q5: Create a multilevel inheritance hierarchy:

- **LivingBeing** (base class): attributes - name, age; methods - breathe(), grow()
- **Animal** (inherits from LivingBeing): adds - species, habitat; methods - move(), eat()
- **Mammal** (inherits from Animal): adds - fur_color, body_temperature; methods - give_birth(), nurse_young()